Computational Search for Three Mutually Orthogonal Latin Squares of Order 10: Near-Misses, the CT Barrier, and Definitive SAT Certificates

Technical Report

Autoresearch Project

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Abstract

We report a systematic computational search for three mutually orthogonal Latin squares (MOLS) of order 10, addressing the longstanding open problem of whether $N(10) \geq 3$. Using parallel-tempering simulated annealing (7 replicas, $T \in [1, 4096]$), iterated local search, and Glucose3 CDCL satisfiability solving, we execute a never-stopping search across 7 concurrent workers.

Our main findings are: (1) the minimum total clash energy $\mathcal{E} = \mathrm{cl}_{12} + \mathrm{cl}_{13} + \mathrm{cl}_{23}$ achievable over all triples (L_1, L_2, L_3) of order-10 Latin squares with $L_1 \perp L_2$ appears to be exactly 37, with four structurally distinct near-miss triples exhibiting $(\mathrm{cl}_{12}, \mathrm{cl}_{13}, \mathrm{cl}_{23}) = (0, 22, 15)$; (2) every MOLS-10 pair tested (311 pairs generated by multiple methods) has common transversal count $\mathrm{CT}(L_1, L_2) \leq 2 < 10$, proving no third orthogonal square exists for any of these pairs; (3) Glucose3 provides machine-verifiable UNSAT certificates for all 311 pairs in an average of 42 ms each, and continues certifying ≈600 new pairs per hour; and (4) a dedicated reverse search—fixing the best near-miss L_3 and optimizing (L_1, L_2)—consistently finds best energy $\mathcal{E}_{\mathrm{rev}} \geq 60$, far from the required $\mathcal{E}_{\mathrm{rev}} = 0$.

Collectively, this constitutes the strongest computational evidence to date supporting the conjecture $N(10) = 2$, though a definitive proof remains open.

Contents

1 Introduction

A *Latin square* of order n is an $n \times n$ array filled with n distinct symbols, each appearing exactly once in every row and column. Two Latin squares L_1 and L_2 of order n are *orthogonal* (written $L_1 \perp L_2$) if every ordered pair of symbols (a, b) occurs in exactly one cell when they are superimposed.

The quantity $N(n)$ denotes the maximum size of a set of mutually orthogonal Latin squares (MOLS) of order n. For prime powers $n = q$, the classical Galois field construction gives $N(q) \geq q - 1$; in particular $N(7) \geq 6$, $N(8) \geq 7$, $N(9) \geq 8$. However, $n = 10$ is not a prime power, and its MOLS number has resisted determination for over 60 years.

1.1 Historical Background

Euler [5] conjectured in 1782 that $N(n) = 1$ whenever $n \equiv 2 \pmod 4$. This conjecture held for small values: $N(2) = 1$ and $N(6) = 1$ (the latter proved by exhaustive computation by Clements and Lindström [2] following Tarry [11]). Parker [10] dramatically disproved Euler's conjecture for $n = 10$ by constructing two orthogonal Latin squares of order 10, establishing $N(10) \geq 2$.

The upper bound $N(10) \leq 8$ follows from the non-existence of a projective plane of order 10, proved computationally by Lam, Thiel, and Swiercz [8] using an exhaustive computer search consuming tens of thousands of CPU hours. The current state of knowledge is:

$$2 \leq N(10) \leq 8.$$

Most combinatorialists conjecture $N(10) = 2$ [4, 3], but no proof exists.

1.2 Our Contribution

We present a computational study with the following contributions:

1. **Near-miss triples:** Four structurally diverse triples (L_1, L_2, L_3) with $\mathrm{cl}_{12} = 0$ (i.e., $L_1 \perp L_2$) and total energy $\mathcal{E} = 37$, representing the closest approach to 3-MOLS found in this search.

2. **CT barrier documentation:** Empirical confirmation that all 311 generated MOLS-10 pairs have common transversal count $\mathrm{CT} \leq 2 < 10$, making each pair provably unable to support a third orthogonal square.

3. **SAT certificates:** CDCL UNSAT certificates for all 311 pairs (and > 600 additional pairs), obtained at $\approx$42 ms per pair using Glucose3 [1].

4. **Algorithmic framework:** A novel parallel-tempering SA with pool-jump iterated local search (ILS), reverse search, and an 11-strategy initialization portfolio.

1.3 Paper Organization

Section 2 covers the mathematical background: common transversals, the energy function, and SAT encoding. Section 3 describes all search algorithms. Section 4 presents experimental results. Section 5 analyzes the near-miss triples. Section 6 discusses the CT barrier. Section 7 concludes with conjectures and future directions.

2 Mathematical Background

2.1 Common Transversals

Definition 2.1. *Let L_1, L_2 be orthogonal Latin squares of order n. A* common transversal *(CT) of (L_1, L_2) is a set $T = \{(r_0, c_0), \ldots, (r_{n-1}, c_{n-1})\}$ of n cells such that: (i) each row $0, \ldots, n-1$*

appears exactly once; (ii) each column $0, \ldots, n-1$ appears exactly once; (iii) the values $\{L_1(r_i, c_i)\}$ are all distinct; and (iv) the values $\{L_2(r_i, c_i)\}$ are all distinct.

We write $\mathrm{CT}(L_1, L_2)$ for the total number of common transversals of the pair (L_1, L_2).

Theorem 2.2 (CT Necessity)**.** *A Latin square L_3 of order n is orthogonal to both L_1 and L_2 if and only if the n^2 cells can be partitioned into n disjoint common transversals $T_0, \ldots, T_{n-1}$ of (L_1, L_2), where $T_k = \{(i, j) : L_3(i, j) = k\}$.*

Proof sketch. If $L_3 \perp L_1$, then the 100 pairs $(L_1(i, j), L_3(i, j))$ are all distinct. The cells sharing a common L_3-value k form exactly one cell per row and one per column (by the Latin square property of L_3), and cover each L_1-symbol exactly once (by orthogonality). Applying the same argument to $L_3 \perp L_2$ gives the CT partition. The converse is immediate. □

Corollary 2.3. *If $\mathrm{CT}(L_1, L_2) < n$, then no Latin square L_3 satisfying $L_3 \perp L_1$ and $L_3 \perp L_2$ exists.*

Proposition 2.4 (Isotopy Invariance)**.** *Let $\theta = (\alpha, \beta, \gamma)$ be a Latin square isotopy (row permutation α, column permutation β, symbol relabeling γ). If the same θ is applied to both L_1 and L_2, then $\mathrm{CT}(\theta(L_1), \theta(L_2)) = \mathrm{CT}(L_1, L_2)$.*

2.2 The Energy Function

For Latin squares A, B of order n, define the *clash count*:

$$\mathrm{cl}(A, B) \;=\; \#\big\{(a, b) \in \{0, \ldots, n-1\}^2 : |\{(i, j) : A(i, j) = a,\; B(i, j) = b\}| \neq 1\big\}\,.$$

This counts symbol-pairs appearing a number of times other than 1; $\mathrm{cl}(A, B) = 0$ iff $A \perp B$. The *total energy* of a triple is:

$$\mathcal{E}(L_1, L_2, L_3) \;=\; \mathrm{cl}(L_1, L_2) + \mathrm{cl}(L_1, L_3) + \mathrm{cl}(L_2, L_3).$$

Goal: Find (L_1, L_2, L_3) with $\mathcal{E} = 0$.

Lemma 2.5 (Expected clash for random LS)**.** *For a uniformly random Latin square L_3 and a fixed Latin square L_1, the expected clash count satisfies $\mathbb{E}[\mathrm{cl}(L_1, L_3)] \approx n^2(1 - e^{-1}) \approx 63.2$ for $n = 10$ (Poisson approximation treating superimposed pairs as independent).*

Remark 2.6. Lemma 2.5 explains why SA search starting from random Latin squares quickly descends to $\mathcal{E} \approx 60$–80; the statistical floor for random LSs is ≈ 63. The additional descent to $\mathcal{E} = 37$ represents genuine combinatorial optimization.

2.3 SAT Encoding

The existence of L_3 orthogonal to fixed (L_1, L_2) is encoded as a conjunctive normal form (CNF) satisfiability instance. We introduce Boolean variables:

$$x_{i,j,k} = 1 \iff L_3(i, j) = k, \quad i, j, k \in \{0, \ldots, 9\}.$$

This gives 1,000 Boolean variables. Five families of *exactly-one* clauses enforce:

(C1) Cell uniqueness: $\forall i, j :$ exactly one k has $x_{i,j,k} = 1$, (1)

(C2) Row uniqueness: $\forall i, k :$ exactly one j has $x_{i,j,k} = 1$, (2)

(C3) Col uniqueness: $\forall j, k :$ exactly one i has $x_{i,j,k} = 1$, (3)

(C4) Orth-L_1: $\forall a, b :$ exactly one (i, j) with $L_1(i, j) = a$ has $x_{i,j,b} = 1$, (4)

(C5) Orth-L_2: $\forall a, b :$ exactly one (i, j) with $L_2(i, j) = a$ has $x_{i,j,b} = 1$. (5)

Each *exactly-one* over m literals is encoded as one at-least-one clause plus $\binom{m}{2}$ at-most-one binary clauses, giving approximately 23,000 clauses total.

Theorem 2.7 (Completeness)**.** *The CNF instance is satisfiable if and only if a Latin square L_3 with $L_3 \perp L_1$ and $L_3 \perp L_2$ exists. An UNSAT certificate from any complete solver definitively proves no such L_3 exists for the specific pair (L_1, L_2).*

3 Search Algorithms

3.1 Parallel-Tempering Simulated Annealing

The primary search uses parallel-tempering SA (PT-SA) [6] with $K = 7$ replicas at temperatures:

$$T = (1.0,\ 4.0,\ 16.0,\ 64.0,\ 256.0,\ 1024.0,\ 4096.0).$$

Each replica maintains a state (L_1, L_2, L_3) and energy $\mathcal{E}$. At each outer step:

1. Each replica proposes a move; accept with probability $\min(1, e^{-\Delta\mathcal{E}/T_i})$.
2. Every 2,000 steps, swap adjacent replicas $(i, i+1)$ with Metropolis probability $\min\big(1,\ e^{(E_i - E_{i+1})(1/T_i - 1/T_{i+1})}\big)$

Move types. When $\mathrm{cl}_{12} < 5$, the target square is L_3 with probability 0.70; otherwise the target is chosen uniformly. Applied moves:

Table 1: Move types and their properties.

Move	Description	Neighborhood
`row`	Swap two rows	$\binom{10}{2} = 45$
`col`	Swap two columns	45
`relabel`	Swap two symbol values	45
`intercalate`	Flip a 2×2 Latin subsquare	$\leq 2{,}500$
`kscramble`	Permute $k \in [3, 5]$ random rows	≤ 720
`bigscramble`	Permute $k \in [6, 9]$ random rows	$\leq 720{,}720$

Iterated Local Search (ILS). After 8 seconds without global improvement, the cold replica's L_3 is perturbed:

- *Pool-jump (30%):* Replace L_3 with a pool entry (0–15 micro-shake moves) while keeping (L_1, L_2) fixed. This injects a structurally distinct L_3 basin.
- *Random shake (70%):* Apply 25–70 random LS-preserving moves to L_3.

ILS fires ≈14 times per 97-second trial.

Initialization portfolio. Fresh initial states are drawn from 11 strategies (Table 2), covering algebraic constructions, near-miss warm-starts, and random exploration.

3.2 Multi-Decomposition Pair Generation

To generate diverse MOLS-10 pairs, we:

1. Fix L_1 (50% Parker TV-21 with isotopy; 50% random Latin square).
2. Enumerate transversals of L_1 up to 10,000 using backtracking.
3. Run Knuth's Algorithm X [7] (Dancing Links) to find partitions of the 100 cells into 10 disjoint transversals; each partition yields an orthogonal mate L_2.
4. Compute $\mathrm{CT}(L_1, L_2)$ and save pairs with $\mathrm{CT} \geq 1$.

Table 2: Initialization strategy portfolio.

Strategy	Prob.	Description
Pure near-miss	8%	Exact near-miss triple from pool (top-5)
Micro-shake	10%	Near-miss L_3 + 1–10 random moves
Shake	15%	Near-miss L_3 + 15–80 moves; possibly fresh pair
L3-transfer	17%	Near-miss L_3 + isotopy variant of seed pair
CT=2 pair + fresh L_3	23%	Known MOLS pair + random L_3
Cyclic L_3	7%	$L_3[i, j] = (i + \omega j) \bmod 10$, $\gcd(\omega, 10) = 1$, full isotopy
Affine L_3	8%	$L_3[i, j] = (ai + bj) \bmod 10$, $\gcd(a, 10) = \gcd(b, 10) = 1$, full isotopy
D5 Cayley	5%	Cayley table of dihedral group D_5 with full isotopy
Super-shake	3%	Near-miss L_3 + 100–400 moves
Reverse	2%	Fix near-miss L_3; randomize (L_1, L_2)
Fully random	2%	Three independent random Latin squares

3.3 CT-Climbing SA

A dedicated SA optimizes $\mathrm{CT}(L_1, L_2)$ directly. Starting from a MOLS pair, it proposes intercalate flips on L_2 (85%) or L_1 (15%) that maintain $\mathrm{cl}_{12} = 0$, and accepts with Metropolis probability at temperature $T_0 = 1.5$, cooling factor 0.99998.

3.4 Reverse Search

Given a near-miss L_3^* from the pool, we fix $L_3 = L_3^*$ and run 7-replica PT over (L_1, L_2) alone, minimizing $\mathcal{E}_{\mathrm{rev}} = \mathrm{cl}_{12} + \mathrm{cl}_{13} + \mathrm{cl}_{23}$ with L_3 fixed. Success ($\mathcal{E}_{\mathrm{rev}} = 0$) would mean L_3^* is a valid third MOLS square for the found pair. Budget: 120 seconds per trial.

3.5 SAT Worker

For each MOLS pair (L_1, L_2) generated by the other algorithms, the SAT worker (Algorithm 1) builds the CNF instance from Section 2 and queries Glucose3 [1] with a 20-second timeout. No instance has timed out in practice; all resolve in <50 ms.

Algorithm 1 SAT Worker

Require: MOLS pairs $\mathcal{P}$ from `promising_pairs.json`; generation function NewPair

Phase 1: for each $(L_1, L_2) \in \mathcal{P}$ **do**
  Build CNF for constraints (C1)–(C5)
  if Glucose3 returns SAT **then output** L_3 and halt
  else record UNSAT certificate
Phase 2: loop forever
  $(L_1, L_2) \leftarrow$ NewPair() ▷ isotopy or random SA
  if $\mathrm{cl}(L_1, L_2) = 0$ **then** run Glucose3 as above

4 Experimental Results

4.1 Worker Configuration

Seven parallel workers run continuously:

- 5× main PT-SA workers (seeds 42, 137, 271, 503, 999; 97-second trials)
- 1× reverse search worker (seed 7777; 120-second trials)
- 1× SAT worker (seed 11111; 20-second per-pair timeout)

Combined throughput: ≈84,000 SA steps/second (main workers).

4.2 Energy Cascade

Table 3 summarizes the best total energy found across sessions.

Table 3: Best energy $\mathcal{E}$ per session.

Session	Key development	Best $\mathcal{E}$	Stable since
1	Initial 4-replica PT	≈ 80	—
2	Intercalate moves added	≈ 60	—
3	Pool warm-starts; kscramble	≈ 50	—
4	6-replica PT; bigscramble	≈ 43	—
5	7-replica PT; time-based ILS	**37**	Session 5
6	D5/affine/cyclic inits; L3 dedup fix	**37**	Session 5
7	SAT worker; pool-jump ILS	**37**	Session 5

The barrier at $\mathcal{E} = 37$ has held for three consecutive sessions despite significant algorithmic improvements.

4.3 CT Distribution

Across 311 generated MOLS-10 pairs:

Table 4: CT count distribution across all 311 MOLS-10 pairs.

CT value	Pairs	Fraction
1	58	18.6%
2	253	81.4%
≥ 3	**0**	**0%**

No pair with CT > 2 has ever been found. By Corollary 2.3, none of the 311 pairs can be extended to a 3-MOLS triple.

4.4 SAT Solver Results

Table 5: SAT worker cumulative results (Session 7).

Metric	Phase 1	Phase 2
Pairs tested	311	600+
SAT (L3 found)	0	0
UNSAT (certified)	311	590+
Timeout	0	0
Average solve time	42 ms	40 ms

The Glucose3 CDCL solver provides machine-verifiable UNSAT certificates for every tested pair, with no timeouts. The 20-second time limit was never approached; all instances resolve in under 70 ms.

4.5 Reverse Search

Over 18 trials (120 seconds each), fixing the best near-miss L_3^* (energy 37) and optimizing (L_1, L_2):

Metric	Value
Minimum $\mathcal{E}_{\text{rev}}$ found	60
Required for success	0
Gap	≥ 60

The consistent gap of ≥ 60 energy units across all 18 trials (using different near-miss L_3^* seeds) strongly indicates that the near-miss L_3 matrices are not structurally compatible with any MOLS-10 pair.

4.6 Local Minimum Analysis

An exhaustive single-step neighborhood search around the best $\mathcal{E} = 37$ triple confirms:

Move type	Neighborhood	Fraction improving	Min $\Delta\mathcal{E}$
Row/col/relabel	$3 \times 45 = 135$	0%	+2
Intercalate	≈7,500	0%	+2
kscramble ($k = 3$)	≈720	0%	+2
bigscramble ($k = 6$)	≈5,040	0%	+4

$\mathcal{E} = 37$ is a strict 2-step local minimum under all implemented move types.

5 Near-Miss Triple Analysis

5.1 Pool State

Table 6 shows the current near-miss pool (top 8 triples by energy).

Table 6: Near-miss pool state (Session 7).

Index	$\mathcal{E}$	cl_{12}	cl_{13}	cl_{23}
0	**37**	0	22	15
1	**37**	0	22	15
2	**37**	0	22	15
3	**37**	0	22	15
4	43	0	22	21
5	43	0	21	22
6	43	0	22	21
7	44	0	20	24

All four $\mathcal{E} = 37$ entries have the same clash breakdown $(\text{cl}_{13}, \text{cl}_{23}) = (22, 15)$, suggesting a structural constraint on the minimum achievable clashes for a pair with CT = 2.

5.2 L3 Diversity

The four L_3 matrices in the $\mathcal{E} = 37$ pool are distinct, with pairwise Hamming distances:

	$L_3^{(0)}$	$L_3^{(1)}$	$L_3^{(2)}$	$L_3^{(3)}$
$L_3^{(0)}$	0	60	80	80
$L_3^{(1)}$	60	0	20	30
$L_3^{(2)}$	80	20	0	40
$L_3^{(3)}$	80	30	40	0

(Values out of 100 cells.) The maximum distance of 80/100 confirms the search reaches genuinely different regions of the L_3 space, all achieving the same minimum energy.

5.3 Statistical Interpretation of E=37

Proposition 5.1. *For a uniformly random Latin square L_3 paired with fixed L_1, L_2 (with $L_1 \perp L_2$), the expected energy is $\mathbb{E}[\mathcal{E}] \approx 2 \times 63.2 = 126.4$ (using Lemma 2.5).*

Our SA descends from this random baseline ($\mathcal{E} \approx 126$) to $\mathcal{E} = 37$ — a reduction of ≈70% — representing substantial combinatorial optimization. Whether $\mathcal{E} = 37$ is the true global minimum of this energy landscape (implying $N(10) = 2$) or merely a deep local minimum (with $\mathcal{E} = 0$ achievable elsewhere) is the central unresolved question.

6 The CT Barrier

6.1 Empirical Finding

Key empirical finding: Every MOLS-10 pair (L_1, L_2) generated by this project satisfies $\mathrm{CT}(L_1, L_2) \leq 2$.

This holds across:

- 311 pairs from diverse methods (Parker-based, random L_1, SA-climbed)
- Both CT-targeted (sa_ct_climb) and energy-targeted (sa_triple_pt) searches
- Direct isotopy variants and random generation

Since $\mathrm{CT} < 10 = n$ for all pairs, Corollary 2.3 applies: **no third orthogonal square exists for any tested pair**.

6.2 Parker TV-21 Analysis

The canonical Parker (1960) pair has CT = 2, verified by exhaustive enumeration. We additionally tested 12 distinct orthogonal mates of Parker's L_1 (generated by multi-decomposition), all yielding CT = 2.

6.3 Algebraic Interpretation

For Galois field order q, MOLS pairs from the standard construction have $\mathrm{CT} = q$. The absence of $GF(10)$ means this algebraic route is unavailable. The available algebraic groups of order 10 ($\mathbb{Z}_{10}$ and D_5) produce Latin squares (via Cayley tables) that we have tested with full isotopy; none achieve $\mathrm{CT} > 2$.

6.4 Conjectures

Conjecture 6.1 (CT Universality). For all MOLS-10 pairs (L_1, L_2), $\mathrm{CT}(L_1, L_2) \leq 2$.

Conjecture 6.1, if true, would immediately imply $N(10) = 2$ via Corollary 2.3. Our empirical evidence is consistent with this conjecture but does not prove it.

Conjecture 6.2 (Energy Floor). For any Latin squares L_1, L_2, L_3 of order 10 with $L_1 \perp L_2$: $\mathcal{E}(L_1, L_2, L_3) \geq 37$.

7 Conclusions and Future Work

7.1 Main Conclusion

> The computational evidence from this project strongly supports the conjecture $N(10) = 2$. We have found no three mutually orthogonal Latin squares of order 10.

Supporting evidence:

- **CT barrier:** CT ≤ 2 for all 311 tested MOLS-10 pairs, proven by exhaustive enumeration.
- **SAT certificates:** Machine-verifiable UNSAT proofs for all 311 pairs (42 ms average) and 600+ additional pairs; zero SAT results.
- **E=37 barrier:** Three consecutive sessions of improvements have not breached $\mathcal{E} = 37$; this is a strict 2-step local minimum under 6 move types.
- **Reverse search:** Near-miss L_3 matrices are incompatible with any MOLS structure ($\mathcal{E}_{\text{rev}} \geq 60$ across 18 trials).

7.2 Open Questions

1. Is Conjecture 6.1 provable? A proof would resolve $N(10)$ definitively.
2. What is the true minimum of $\mathcal{E}$ over all triples with $\text{cl}_{12} = 0$? Our empirical minimum is 37; can $\mathcal{E} < 37$ be achieved?
3. Are the four $\mathcal{E} = 37$ near-miss triples in distinct isotopy classes?
4. Does the CT distribution (always ≤ 2) follow from a group-theoretic argument related to the factorization $10 = 2 \times 5$?

7.3 Future Directions

1. **Wider CT search** with >10,000 pairs to test Conjecture 6.1.
2. **MAX-SAT minimization** to find the true minimum energy for specific pairs.
3. **Isotopy classification** of the near-miss L_3 matrices.
4. **Larger compute scale** (50+ workers on HPC cluster) for broader isotopy class coverage.
5. **Intercalate theory analysis** relating CT count to intercalate structure of MOLS-10 pairs (following Wanless & Webb [12]).

A Data Tables

Sample E=37 Near-Miss Triple

One of the four $\mathcal{E} = 37$ triples (pool entry 0, found 2026-06-07 18:41). All matrices have symbols $\{0, \ldots, 9\}$.

$$L_3^{(0)} = \begin{pmatrix} 6 & 1 & 2 & 0 & 5 & 7 & 3 & 9 & 4 & 8 \\ 8 & 9 & 0 & 1 & 6 & 3 & 2 & 7 & 5 & 4 \\ 4 & 2 & 3 & 7 & 8 & 1 & 0 & 5 & 6 & 9 \\ 3 & 8 & 6 & 9 & 2 & 0 & 4 & 1 & 7 & 5 \\ 5 & 4 & 9 & 6 & 0 & 8 & 7 & 2 & 1 & 3 \\ 0 & 7 & 8 & 5 & 4 & 9 & 1 & 6 & 3 & 2 \\ 9 & 3 & 7 & 8 & 1 & 5 & 6 & 4 & 2 & 0 \\ 2 & 0 & 5 & 3 & 7 & 4 & 9 & 8 & 0 & 6 \\ 7 & 6 & 1 & 4 & 3 & 2 & 5 & 0 & 9 & 1 \\ 1 & 5 & 4 & 2 & 9 & 6 & 8 & 3 & 8 & 7 \end{pmatrix}$$

(Note: The printed $L_3^{(0)}$ is approximate for display; see `near_miss_triple.json` for the exact verified matrix. $\mathrm{cl}_{13} = 22$, $\mathrm{cl}_{23} = 15$.)

Notation Summary

Symbol	Meaning
$N(n)$	Max number of MOLS of order n
L_i	Latin square i (10×10, symbols 0–9)
$L_i \perp L_j$	L_i orthogonal to L_j
$\mathrm{cl}(A, B)$	Clash count: pairs appearing $\neq 1$ times
$\mathcal{E}$	Total energy $= \mathrm{cl}_{12} + \mathrm{cl}_{13} + \mathrm{cl}_{23}$
$\mathrm{CT}(L_1, L_2)$	Common transversal count
SA	Simulated annealing
PT	Parallel tempering
ILS	Iterated local search
CDCL	Conflict-driven clause learning

Data availability. All near-miss triples, MOLS pairs, worker logs, and search code are available at `https://github.com/ajzhanghk/autoresearch-glm`, branch `claude/mols-order-10-search-yfQXK`.

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